CertiVar: Prevalence-Free Certified Assay Likelihood Ratios for Missense Variant-Effect Predictor Calls

Aaryav Sharma
Eastview High School, Apple Valley, MN, USA
aaryavvatts@gmail.com

Abstract

Variant-effect predictors are a common source of evidence for missense variants of uncertain significance (VUS), and the Bayesian form of the ACMG/AMP guidelines expresses evidence strength as a likelihood ratio. Gene-specific calibrations estimate this ratio from few labelled variants, as point estimates or resampling-based bounds that are not designed to remain valid once a threshold has been chosen from the same data. We present CertiVar, which turns a predictor score into loss-of-function (LoF) calls, functional calls or abstentions using class-conditional conformal thresholds learned from saturation genome editing or deep mutational scanning data, with one variant per residue. It then certifies a bound on the likelihood ratio of each call type with respect to the assay outcome that stays valid after a threshold is chosen from a grid, by a Bonferroni correction, or certifies evidence levels by fixed-sequence testing. Because these guarantees condition on the assay class, none needs the unknown share of LoF variants among the variants being called, provided their within-class score distribution matches that of the assayed variants. On 14,477 assayed missense variants in BRCA1, BRCA2, BAP1, RAD51C and MSH2, scored by AlphaMissense, REVEL and ESM-2, worst-class coverage was 0.896–0.908 at a 0.90 target in all 15 gene–predictor pairs and stayed at 0.89 or above when the LoF share of the calibration data ranged from 5% to 50%; marginal conformal prediction calibrated at a 5% share covered LoF variants at only 0.11 in its worst pair. In resampling experiments with known population likelihood ratios, all certified bounds of a draw held in 99.8% of draws (guaranteed: at least 90%), against 56.5% for percentile-bootstrap bounds, and fixed-sequence testing, which certifies one evidence level at a time, more than doubled the certification of moderate LoF evidence. The certificates concern the assay, not pathogenicity; certified evidence was mostly supporting-level and held for 88.8% of certified bounds in held-out protein regions, below the nominal 95%. Code: `https://github.com/aaryavvatts-lab/certivar`

Keywords: variant effect prediction; conformal prediction; likelihood ratio; ACMG/AMP evidence calibration; saturation genome editing; deep mutational scanning; variants of uncertain significance; multiple testing.

1 Introduction

In the five cancer-predisposition genes studied here, 81% of the ClinVar missense records classified as (likely) pathogenic, (likely) benign or uncertain are variants of uncertain significance (VUS; our count from the 26 October 2026 release [1]), and for many of them computational predictors are among the few sources of evidence [2]. The guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) grade evidence qualitatively [3]; their Bayesian adaptation equates each strength with a likelihood ratio, the odds of pathogenicity (OddsPath) [4, 5]. Calibration workflows report this ratio as a point estimate or a resampling-based bound [2, 6, 7]. For a single gene, few labelled variants are available and the threshold that defines the evidence is chosen from the same data; these bounds are not designed to remain valid after this choice.

Two features of the data make the estimate fragile. First, the share of damaging variants differs between calibration sets and the variants being classified, and is unknown for the latter: in our genes, 45% of confidently labelled ClinVar missense variants are pathogenic, whereas 6.0% to 25.9% of the other assayed missense variants lose function in saturation genome editing (SGE) or deep mutational scanning (DMS) experiments (Table 1). Second, gene-level labelled sets are small and substitutions at one residue are dependent: in a resampling design with known truth, percentile-bootstrap bounds were respected in only 76.5% of threshold selections (Section 4.4).

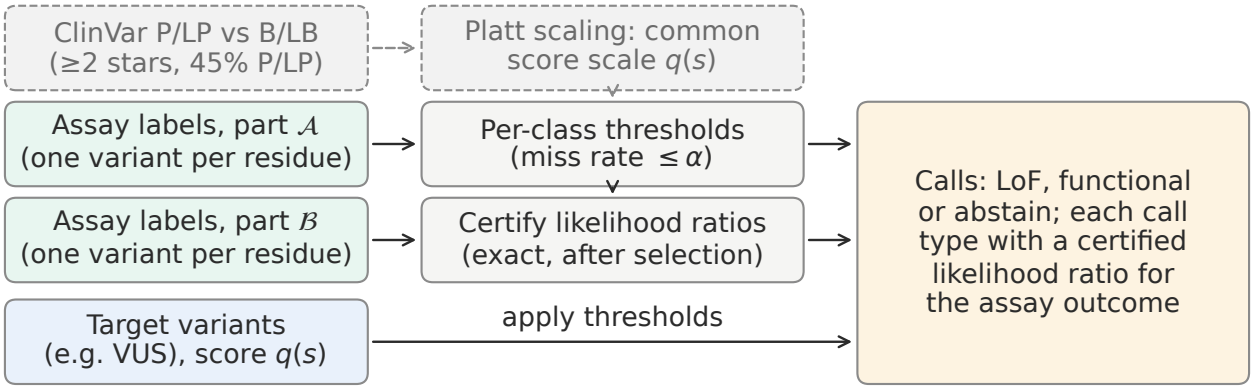


Figure 1: CertiVar. Per-class thresholds are set on part $\mathcal{A}$ of a gene's assayed residues, the likelihood ratio of each call type with respect to the assay outcome is certified on a disjoint part $\mathcal{B}$, and the thresholds are applied to the target variants. The grey path (Platt scaling on ClinVar labels) only puts predictors on a common scale and does not change any call.

We introduce CertiVar (Figure 1), which turns a predictor score into loss-of-function (LoF), functional or no calls. Per-class (Mondrian) conformal thresholds set from SGE or DMS labels bound how often each class is mis-called, and a certification step bounds the likelihood ratio of each call type with respect to the assay outcome, valid after a threshold has been selected; neither needs an estimate of the LoF share. The certified quantity is how strongly a call predicts the assay result; it is evidence about pathogenicity only to the extent that the assay is clinically validated. Our contributions: (1) a formulation of predictor-based calling as abstaining classification on residue-clustered data, with proofs that the certified likelihood-ratio bounds hold after threshold selection, that per-class coverage after selection is at least one minus the largest grid level, and that a fixed-sequence variant needs no multiplicity correction over the grid (Section 3); (2) a known-truth benchmark in which exact binomial bounds held where bootstrap bounds and point estimates did not (Section 4.4); (3) evidence on five genes and three predictors that per-class coverage stays at or near the target when the calibration LoF share moves from 5% to 50%, where marginal and prior-reweighted conformal prediction fail (Section 4.2); and (4) where the guarantees stop (Section 4.5).

2 Related work

ClinGen maps predictor scores to ACMG/AMP evidence strengths through local likelihood ratios estimated against ClinVar labels, with a one-sided bootstrap bound [2]; the approach has been extended to more predictors [6], to assay scores [5, 8, 9] and to gene- and domain-specific calibration [7, 10, 11], with uncertainty from the bootstrap [2, 6] or sign-test bounds on resampled likelihood ratios [7]. Split conformal prediction gives finite-sample coverage [12]; its class-conditional (Mondrian) form controls coverage within each class [12, 13]; order-statistic thresholds give training-conditional control [14, 15]; keeping one member per group handles clustered data [16]; and abstaining classifiers trade the share of inputs classified against error [17]. Selecting a threshold from a grid with binomial bounds and a multiplicity correction, or by fixed-sequence testing, is the template of selective classification with guaranteed risk, risk-controlling prediction sets and Learn-then-Test [18–20]. Harrizi et al. reported split-conformal prediction sets for missense predictors on ClinVar benchmarks at 90% coverage [21]; CertiVar instead controls coverage within each class, calibrates on the gene's assay labels and certifies the likelihood ratio of the calls. Large-sample intervals for a ratio of binomial proportions [22, 23] are narrower than our ratio of exact endpoints but carry no finite-sample guarantee. Label-shift methods adjust posteriors or conformal weights to an estimated target prior [24–26]; class-conditional quantities need no prior, provided the target satisfies class-conditional invariance (Assumption 2). Relative to these methods, CertiVar contributes a certificate on the OddsPath scale for both call types of an abstaining rule on residue-clustered assay labels, with coverage after threshold selection, a fixed-sequence variant, and a known-truth benchmark of how often point estimates and bootstrap bounds overstate evidence.

3 Method

Setting. Each missense variant has a predictor score s, oriented so that higher means more damaging, and an assay label $Y \in \{0, 1\}$, with $Y = 1$ for loss of function (LoF). A *residue* is a protein position of one gene; the substitutions at a residue share structural context and form one cluster. A gene's assayed missense variants are split by residue into a calibration part $\mathcal{A}$ and a certification part $\mathcal{B}$; the variants to be called (for example VUS) form the target. We write $q(s) \in [0, 1]$ for the score after Platt scaling [27] fitted to confidently labelled ClinVar variants (pathogenic or likely pathogenic, P/LP, against benign or likely benign, B/LB); $q(s)$ estimates the probability that such a ClinVar variant is pathogenic, not that an assay outcome is LoF. Because q is increasing in s, every procedure below makes the same calls on q as on s; q only puts predictors on one scale (the baselines use its values), so no ClinVar label affects a CertiVar call.

Per-class thresholds. The nonconformity of label 1 is $r_1 = 1 - q$ and of label 0 is $r_0 = q$. From each residue in $\mathcal{A}$ we keep one variant chosen uniformly at random. For class k, let n_k be the number of kept variants with label k and $r^{(k)}_{(1)} \le \cdots \le r^{(k)}_{(n_k)}$ their sorted nonconformities. The threshold is $\hat{t}_k = r^{(k)}_{(j_k)}$ (∞ if $j_k > n_k$), where j_k is the in-expectation rank or the probably approximately correct (PAC) rank ($j_k = n_k + 1$ if no j qualifies):

$$j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil \text{ (in expectation)} \quad \text{or} \quad j_k = \min\{\, j : \; F_{\mathrm{Beta}(j,\, n_k - j + 1)}(1 - \alpha_k) \le \delta \,\} \text{ (PAC).} \tag{1}$$

The prediction set is $C(q) = \{k : r_k \le \hat{t}_k\}$; CertiVar calls LoF if $C = \{1\}$, functional if $C = \{0\}$, and abstains otherwise. A functional call on a LoF variant requires $1 \notin C$, so per-class coverage bounds both error rates.

Certified likelihood ratios. The ACMG/AMP guidelines [3] grade evidence in qualitative categories; in their Bayesian adaptation, supporting, moderate, strong and very strong evidence towards pathogenicity correspond to a likelihood ratio above 2.1, 4.3, 18.7 and 350, and supporting, moderate and strong evidence towards benignity to one below 0.48, 0.23 and 0.053 [4, 5]. We use these values as levels and write, for example, *LoF-supporting* for LoF calls whose likelihood ratio exceeds 2.1. Our likelihood ratios are defined with respect to the assay outcome Y and averaged over all variants that receive a call, as OddsPath is for a functional assay [5]; they are not the local, score-specific ratios that ClinGen uses for the computational criteria PP3/BP4 [2]. A level thus describes the call class as a whole, need not hold for each call, and must not be combined with the same assay's functional evidence (PS3/BS3) or with PP3/BP4 from the same predictor. For thresholds t, the call rates (probabilities, not counts) are $\mathrm{TP}(t) = \Pr(\text{call LoF} \mid Y = 1)$, $\mathrm{FP}(t) = \Pr(\text{call LoF} \mid Y = 0)$, $\mathrm{FN}(t) = \Pr(\text{call functional} \mid Y = 1)$ and $\mathrm{TN}(t) = \Pr(\text{call functional} \mid Y = 0)$, and $\mathrm{LR}^+ = \mathrm{TP}/\mathrm{FP}$, $\mathrm{LR}^- = \mathrm{FN}/\mathrm{TN}$.

For each α on a grid G we compute thresholds on $\mathcal{A}$ ($\alpha_0 = \alpha_1 = \alpha$), call one variant per residue of $\mathcal{B}$, and bound the four rates by one-sided Clopper–Pearson bounds [28] at level $\epsilon = \delta/(2|G|)$, written ℓ (lower) and u (upper): for x calls among n variants of a class, $\ell = F^{-1}_{\mathrm{Beta}(x,\, n-x+1)}(\epsilon)$ (0 if $x = 0$) and $u = F^{-1}_{\mathrm{Beta}(x+1,\, n-x)}(1 - \epsilon)$ (1 if $x = n$, including $n = 0$). The certified bounds are $L^+(\alpha) = \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$ and $U^-(\alpha) = u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ ($U^- = \infty$, never meeting a level, if $\ell_{\mathrm{TN}} = 0$; $u_{\mathrm{FP}} > 0$ always; for point estimates, $x/0 = \infty$ for $x > 0$, and $0/0$ is undefined and never selected). For each level, Algorithm 1 returns the α that calls the most variants among those whose certified bound meets the level. Because $\hat{t}_0$ and $\hat{t}_1$ both decrease weakly in α, no variant can be called LoF at one α and functional at another (Supplement S1). Line 7 also requires, for functional calls, α at least as large as the LoF choice of the same strength; this is a reporting convention, not needed for validity, and can only lower functional yields. The *fixed-sequence* variant replaces the Bonferroni level $\delta/(2|G|)$ by $\delta/2$ and tests the grid in an order fixed in advance, stopping at the first α that does not meet the level (Theorem 2). We test LoF levels in increasing and functional levels in decreasing α, because few variants are called functional at small α. All thresholds and bounds cost $O(|G|\, n \log n)$ for n residues, about 5 ms per gene on a laptop CPU.

Algorithm 1 CertiVar for one gene and one evidence level

Require: scores q; assay labels split by residue into $\mathcal{A}, \mathcal{B}$; grid G; δ; targets τ (LoF) and τ' (functional)

1: keep one variant per residue of $\mathcal{A}$ and of $\mathcal{B}$
2: **for** $\alpha \in G$ **do**
3: thresholds $\hat{t}_0, \hat{t}_1$ on the kept variants of $\mathcal{A}$ by (1); call the kept variants of $\mathcal{B}$
4: $L^+(\alpha) \leftarrow \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$, $U^-(\alpha) \leftarrow u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ (Clopper–Pearson, level $\delta/(2|G|)$ each)
5: **end for**
6: $\hat{\alpha}^+ \leftarrow \arg\max_{\alpha \in G}\{\text{share called LoF on } \mathcal{B} : L^+(\alpha) > \tau\}$
7: $\hat{\alpha}^- \leftarrow \arg\max_{\alpha \in G}\{\text{share called functional on } \mathcal{B} : U^-(\alpha) < \tau',\ \alpha \geq \hat{\alpha}^+\}$ (over all $\alpha \in G$ if $\hat{\alpha}^+$ does not exist)
8: **output** LoF calls at $\hat{\alpha}^+$ and functional calls at $\hat{\alpha}^-$ with their certified levels (no calls of a type whose set is empty); abstain otherwise

3.1 Guarantees

Let Q be the law of a variant obtained by drawing a residue from the gene's population of assayed residues and then one of its variants uniformly at random (binary-labelled, non-splice variants outside the ClinVar labelled set).

Assumption 1 (Clustered sampling)**.** Residues in $\mathcal{A}$, in $\mathcal{B}$ and of a test variant are independent draws from one residue population; within a residue, the kept or tested variant is chosen uniformly at random; q does not depend on these data.

Assumption 2 (Class-conditional invariance)**.** Target variants have, for $k \in \{0, 1\}$, the same law of q given $Y = k$ as a test variant under Assumption 1; their LoF share $\pi = \Pr(Y = 1)$ may differ.

Proposition 1 (Per-class coverage)**.** *Under Assumption 1, for a fixed α_k, with the in-expectation rank and conditionally on n_k, $1 - \alpha_k \leq \Pr(k \in C \mid Y = k, n_k) \leq 1 - \alpha_k + 1/(n_k + 1)$, the upper bound holding when there are no ties.*

Proposition 2 (High-probability per-class coverage)**.** *Under Assumption 1, for a fixed α_k and with the PAC rank, $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \alpha_k$ with probability at least $1 - \delta$ over the calibration data for each k, and for both classes with probability at least $1 - 2\delta$.*

Proposition 3 (Coverage after selection)**.** *Under Assumption 1, for any $\hat{\alpha} \in G$ chosen from data independent of the test variant (for example by Algorithm 1), the in-expectation rank gives $\Pr(k \in C \mid Y = k) \geq 1 - \max G$; the PAC rank at level $\delta/|G|$ gives $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \hat{\alpha}$ for each k with probability at least $1 - \delta$.*

Theorem 1 (Certified likelihood ratios)**.** *Under Assumption 1, conditionally on $\mathcal{A}$ and with probability at least $1 - \delta$ over $\mathcal{B}$, $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ under Q for every $\alpha \in G$ simultaneously; the same holds for $\mathrm{LR}^- \leq U^-$ with a second δ. Because the event does not depend on the level, each call type returned by Algorithm 1 reaches its certified level at every evidence level reported, with probability at least $1 - \delta$ (both call types with probability at least $1 - 2\delta$), whichever α is selected.*

Theorem 2 (Fixed-sequence certification)**.** *Under Assumption 1, fix a level τ and an order of G chosen in advance. Reject $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ when $\ell_{\mathrm{TP}}/u_{\mathrm{FP}} > \tau$ with both endpoints at level $\delta/2$, stop at the first α not rejected, and select the rejected α (a prefix of the order) with the largest yield. Then, conditionally on $\mathcal{A}$, the selected LoF calls have $\mathrm{LR}^+ > \tau$ with probability at least $1 - \delta$. The same holds for functional calls with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$.*

Corollary 1 (No prior needed)**.** *Under Assumptions 1 and 2, Propositions 1–3 and Theorems 1–2 hold for target variants whatever their LoF share $\pi \in (0, 1)$: for a fixed α, LoF variants are called functional at rate at most α_1, and the certified likelihood ratios with respect to the assay outcome need no prior.*

Propositions 1 and 2 are the class-conditional and training-conditional conformal guarantees [12–15], here for residue-clustered data. Proposition 3 holds because both ranks in (1) are non-increasing in α, so coverage at any selected α is at least coverage at $\max G$. Theorem 1 follows the risk-control template [19]: given $\mathcal{A}$ and the labels in $\mathcal{B}$, the four counts are binomial, and a union bound over the $2|G|$ endpoints covers every α, including a selected one. Theorem 2 follows Learn-then-Test [20]: a true H_α is rejected only if an endpoint fails, and a false claim requires rejecting the first true hypothesis in the order. Proofs are in Supplement S1.

Table 1: Datasets. *ClinVar labelled*: missense single-nucleotide variants (SNVs) with ≥2 review stars and a P/LP or B/LB call (used to fit the common score scale and, in Table S4 only, for an exploratory check) and the number that are P/LP. All shares are variant-level (descriptive). *Assayed*: missense SNVs with a binary assay label that are not in the ClinVar labelled set, and their LoF share; *assayed VUS*: the subset classified VUS in ClinVar. Assay labels are the authors' published classes (authors) or, where none are deposited, derived from the assay's own synonymous and nonsense controls (controls).

			ClinVar labelled		Assayed		Assayed VUS	
Gene	Assay	Labels	n	P/LP	n	LoF	n	LoF
BRCA1	SGE [29]	authors	345	156	1,617	0.161	272	0.173
BRCA2	SGE [30]	controls	309	94	3,435	0.130	704	0.207
BAP1	SGE [31]	authors	14	6	4,294	0.213	1,069	0.166
RAD51C	SGE [32]	authors	13	10	2,225	0.259	550	0.396
MSH2	DMS [33]	authors	178	121	2,906	0.060	1,542	0.067
Total			859	387	14,477		4,137	

Scope. After selection, per-class coverage is guaranteed only at $1-\max G$ with the in-expectation rank; we report the coverage achieved in Section 4.4. Theorem 2 certifies one level at a time and gives no numerical bound. Assumption 1 treats assayed residues as exchangeable, which a contiguous region need not be; Assumption 2 holds only approximately for VUS, on which AlphaMissense discriminates less well (Section 4.1), and not for variants whose assay result would be intermediate.

4 Experiments

4.1 Data and protocol

We took every GRCh38 missense single-nucleotide variant in BRCA1, BRCA2, BAP1, RAD51C and MSH2 that is in ClinVar [1] (release of 26 October 2026) or was assayed, annotated each on its MANE Select transcript with the Ensembl Variant Effect Predictor [34] (AlphaMissense [35], REVEL [36]) and computed ESM-2 650M [37] log-likelihood ratios by wild-type marginals [38]. Labels come from five MaveDB [39] score sets (Table 1; accessions in Supplement S5): the authors' classes for BRCA1, BAP1 and RAD51C, the authors' threshold for MSH2 (positive loss-of-function score [33]), and, for BRCA2, which deposits no classes, Gaussians fitted to the assay's synonymous and nonsense variants (LoF if the posterior is at least 0.95, functional if at most 0.05). Because this control rule disagreed with the authors' classes for 216 of the 4,023 BAP1 variants confident under both, we repeat the main experiments under alternative label rules (Section 4.5). We excluded the 729 assayed variants in splice regions and set aside the 1,274 intermediate variants. One Platt map per predictor, pooled over genes, used the 859 ClinVar missense variants with at least two review stars and a P/LP or B/LB call (45.1% P/LP), which are excluded from every assay-based evaluation set, leaving 14,477 assayed variants with a binary label. Pooled over genes, AlphaMissense has an area under the ROC curve (AUROC) of 0.946 on the ClinVar set, 0.840 on the other assayed variants and 0.779 on assayed VUS (non-overlapping 95% residue-bootstrap intervals). Baselines are pooled (marginal) split conformal prediction; label-shift-weighted conformal prediction [26] with the true target share or its expectation–maximization (EM) estimate [24]; fixed tiers ($q \geq 0.9$ LoF, $q \leq 0.1$ functional); the published AlphaMissense classes (> 0.564, < 0.34); and the REVEL PP3/BP4 thresholds (≥ 0.644, ≤ 0.290) [2]. None claims per-class coverage, and the published cut-offs target pathogenicity, so we judge them only as assay classifiers. Coverage is residue-weighted as in Q; *worst-class coverage* is the smaller of the mean LoF and mean functional coverages over repetitions. Every split is by residue (designs in Supplement S6).

4.2 Per-class coverage, and a shift in the calibration LoF share

Over 50 random halvings of each gene's residues (calibrate on one half, test on the other), the in-expectation rule at $\alpha = 0.1$ gave worst-class coverage between 0.896 and 0.908 in all 15 gene–predictor pairs, none more than one Monte Carlo standard error (0.002–0.010) below the target (Figure 2A; per-pair values in

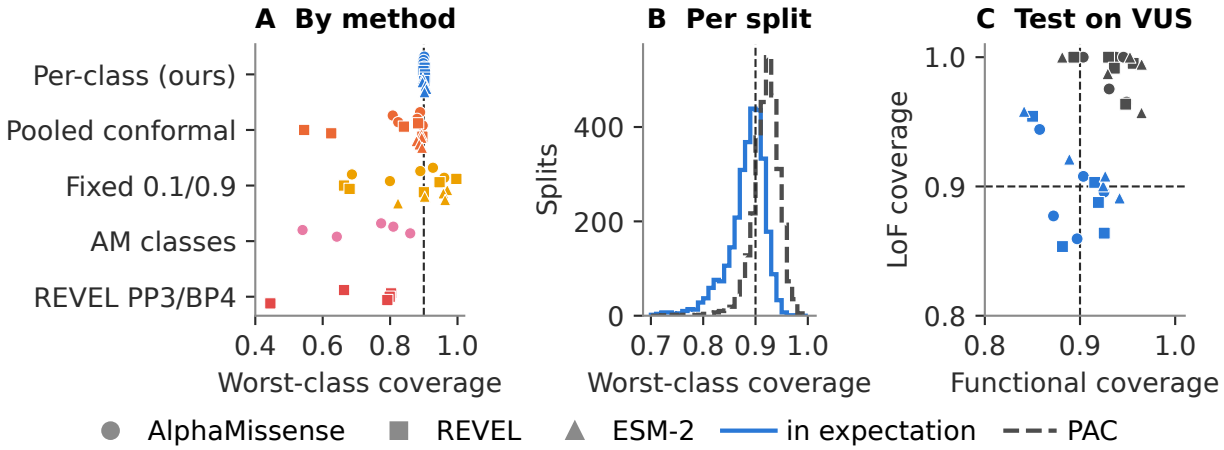


Figure 2: Per-class coverage at $\alpha = 0.1$; markers give the predictor in panels A and C. (A) Worst-class coverage for each gene–predictor pair (50 splits); AM classes: the published AlphaMissense classes. (B) Smaller of the two per-class coverages in each split, pooled over pairs (3,000 splits per rule). (C) Coverage on assayed VUS with thresholds set on non-VUS variants at other residues (50 repetitions); in C, blue: in expectation, grey: PAC.

Table S1). Pooled conformal fell to 0.545 on LoF variants and below 0.85 in 5 pairs, because a marginal threshold controls coverage averaged over the two classes, not within each; the published AlphaMissense and REVEL cut-offs covered as little as 0.540 (BAP1) and 0.445 (MSH2) of functional variants. The per-class rule abstained on 35% of variants on average (17–47%), against 42% for pooled conformal and 11% for the AlphaMissense classes, which trade coverage for decisiveness. Calibrating on all variants instead of one per residue used five times as many labels and abstained less (30.4% against 35.2%), but worst-class coverage fell below 0.90 in 13 of 15 pairs (4 of 15 with one variant per residue).

To test whether per-class thresholds need a prevalence estimate, we subsampled each calibration half to a LoF share of 0.05 to 0.5 and tested at the natural share (100 repetitions; Figure 4A). The per-class rule kept worst-class coverage between 0.897 and 0.925 in every pair at every share (0.891–0.925 in a separate run balanced at 50% LoF), with abstention rising from 37% to 83% as LoF variants became scarce. Pooled conformal calibrated at share 0.05 covered LoF variants at 0.113 in its worst pair, and label-shift-weighted conformal fell to 0.53–0.60 with the true target share and to 0.04 with the EM estimate, which missed the LoF share by 0.255 on average because ClinVar pathogenicity and assay loss of function differ in definition as well as in class share.

4.3 Reliability across splits and a real change of population

With 17 to 93 LoF residues per calibration half, a single split can miss the in-expectation guarantee: over 200 splits per pair, LoF coverage was below 0.90 in 36.4% of splits and functional coverage in 43.6% (Figure 2B). PAC thresholds ($\delta = 0.1$) reduced these to 7.9% and 10.8% (test-half estimates of the coverage that Proposition 2 bounds) at the price of abstaining on 58.4% of variants against 35.4%. Calibrated on non-VUS variants of one half of each gene's residues and applied to the assayed VUS of the other half (residue-weighted LoF share 17.0% in calibration, 19.8% in VUS), in-expectation thresholds kept LoF coverage between 0.854 and 0.958 and functional coverage between 0.841 and 0.942, each below 0.90 in 7 of 15 pairs (Figure 2C); PAC thresholds stayed at or above 0.957 (LoF) and 0.881 (functional) but abstained on 81.7% of VUS.

4.4 Certified likelihood ratios

Treating each gene's assayed residues as the population (the oracle design), we drew calibration (30%) and certification (50%) residues with replacement, 100 times per pair, ran Algorithm 1 with $\delta = 0.05$ and $G_4 = \{0.05, 0.10, 0.20, 0.30\}$, and computed the true residue-weighted likelihood ratio of every selection on the whole population. We compared Clopper–Pearson bounds without multiplicity correction, a one-sided

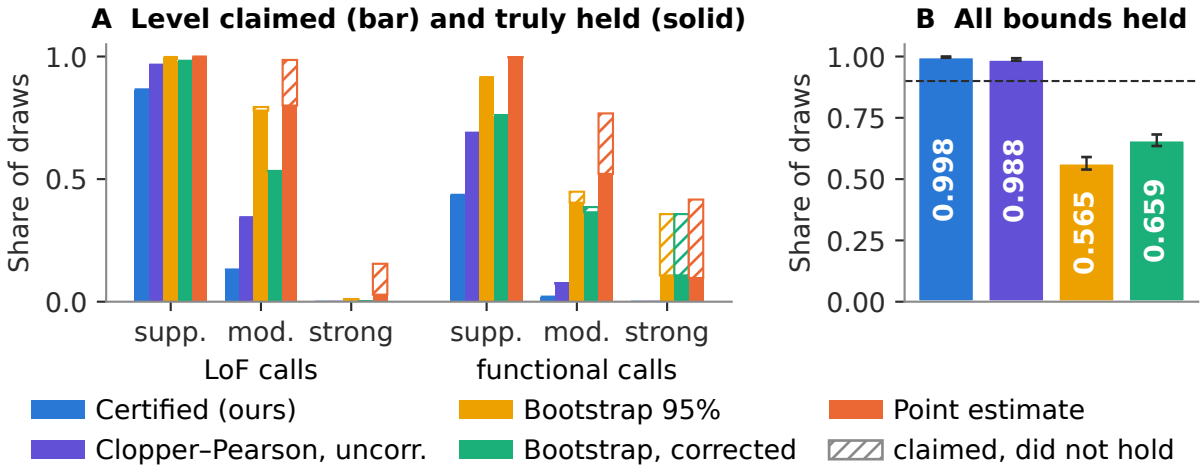


Figure 3: Threshold selection judged against the population likelihood ratio (oracle design, 1,500 draws). (A) Share of draws in which each rule claims a level (full bar) and in which it held (solid). (B) Share of draws in which every claimed bound was respected, with 95% intervals from resampling draws (a point estimate is not a bound and is omitted); dashed line at the guaranteed 0.90.

95% percentile bootstrap (2,000 resamples), the same bootstrap at level $\delta/(2|G|)$, and the point estimate, all applied to the same data. Every bound claimed in a draw was respected in 99.8% of the 1,500 draws for the certified rule (Theorem 1 guarantees at least 90%), 98.8% for uncorrected bounds, 56.5% for bootstrap bounds and 65.9% for the corrected bootstrap (Figure 3B); per selection, the bootstrap bound was respected in only 76.5%. Most bootstrap failures came from certification samples with no false call of that type, where the bound failed 89% of the time. Claimed levels held in all 2,181 certified claims, but in only 91.0% of bootstrap and 79.7% of point-estimate claims (Figure 3A). On this grid the exact bound on one variant per residue, not the multiplicity correction, separates valid rules from the others; the correction costs yield (LoF-moderate certified in 13.1% of draws with it, 34.5% without).

CertiVar certified LoF-supporting calls in 86.6% of draws (calling 25.8% of variants), functional-supporting calls in 43.6% (61.1% called), LoF-moderate in 13.1% and functional-moderate in 2.1%, and never a strong level. In a fresh set of 1,500 draws, per-class coverage at the selected α averaged 0.82 (LoF) and 0.80 (functional), above the guaranteed 0.70 (Proposition 3): selecting for evidence strength trades coverage for confidence. Fixed-sequence testing, which certifies one level at a time, raised LoF-moderate certification in these draws from 12.9% (Bonferroni) to 29.1% of draws and functional-supporting from 42.5% to 60.6% (Figure 4B), with 99.7% of 2,705 claims holding; on a 20-level grid it kept LoF-moderate at 26.7% where the Bonferroni rule fell to 6.2%. The testing order matters: functional-supporting, tested in increasing α, was certified in only 25.7% of draws (60.6% in decreasing order). On held-out fifths of the residues, empirical likelihood ratios respected the certified bounds in 98.7% of 2,091 cases (bootstrap: 76.8%); on VUS at other residues, LoF-supporting held in all 443 certified claims and LoF-moderate in 25 of 28. Against ClinVar labels (exploratory, not independent of the assays), LoF calls had likelihood-ratio lower bounds of 4.9–19.6 (BRCA1), 7.6–9.4 (BRCA2) and 2.6–3.2 (MSH2) (Table S4).

4.5 How much data, from where, and with which labels

Calibration data had to come from the same gene: calibrating on the other four genes left 8 of 15 pairs below 0.85 (worst 0.487), consistent with gene-specific evidence strength [10, 11]. Calibrating on three contiguous quarters of a protein and testing on the fourth gave worst-class coverage of 0.860–0.923, and the true ratio respected 88.8% of 1,891 certified bounds, below the nominal 95%. With 150 calibration residues drawn at random, coverage averaged over genes met the target with 40.5% abstention (82.8% with 25 residues; Fig. S2). Coverage was lower on the third of variants nearest the assay class boundary (0.865 LoF, 0.873 functional). Intermediate variants, which no guarantee covers, were called LoF in 19.6% and functional in 36.6% of cases; on 1,534 unassayed BRCA1 VUS outside the assayed region, AlphaMissense thresholds

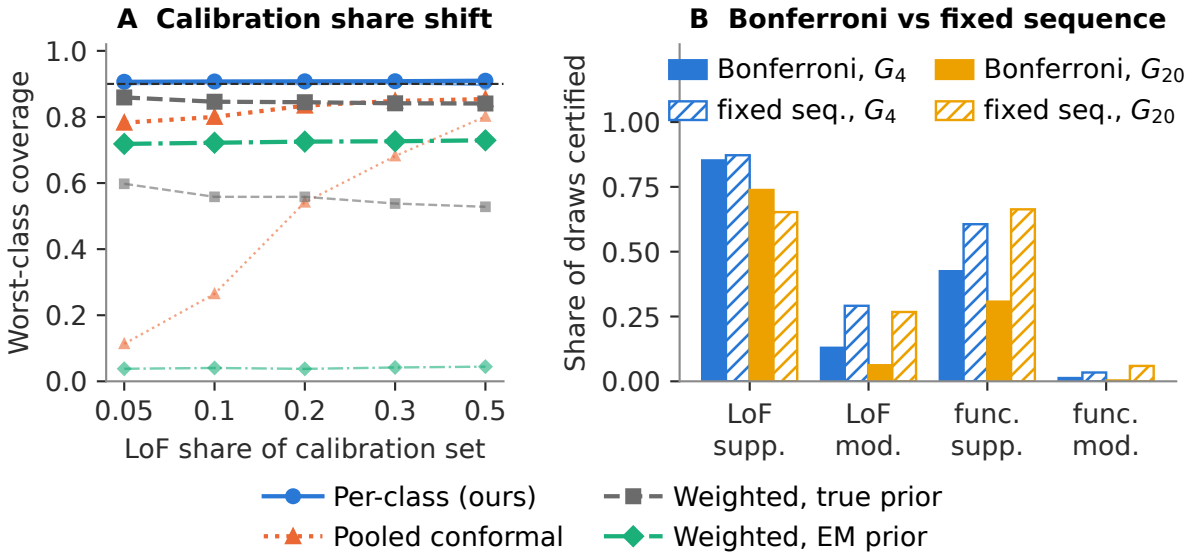


Figure 4: (A) Worst-class coverage when the calibration half is subsampled to a given LoF share (evenly spaced axis): mean over 15 pairs (thick) and worst pair (thin). (B) Share of oracle draws certifying each level with a Bonferroni correction or fixed-sequence testing, on four-level (G_4) and 20-level (G_{20}) grids.

abstained on only 53, which suggests a shifted score distribution. Relabelling BRCA2, BAP1 and RAD51C with alternative rules (Table S3) changed yields but not validity: worst-class coverage stayed between 0.894 and 0.908 over rules and predictors, and certified bounds held in 2,530 of 2,533 oracle selections.

5 Discussion

Thresholds selected on point estimates or percentile-bootstrap bounds often claimed more evidence than held. Because these are call-averaged ratios, not the local ratios of ClinGen calibrations, this does not show that published PP3/BP4 thresholds are wrong, only that point estimates and bootstrap bounds can overstate evidence at single-gene sample sizes. CertiVar makes the claim checkable without an estimate of the share of damaging variants. Its intended use is a gene with a functional assay: calling variants the assay missed within assayed regions, and checking whether a predictor's claimed evidence strength holds against the assay. The cost is conservatism: certified evidence was mostly supporting-level, and PAC thresholds abstained on 58% to 82% of variants.

Limitations. The labels are assay classes, not clinical pathogenicity [5]; clinical use requires an assay validated against clinical classifications, as its authors did for BRCA1 [29] and, with their own classes rather than our control-based labels, for BRCA2 [30]. A certified level describes the calls as a class, so a borderline call can carry less evidence than its class. The guarantees assume residues exchangeable with the calibrated ones; they do not cover intermediate, splice-region or out-of-region variants, and per-class coverage after selection is guaranteed only at $1 - \max G$. Results cover five cancer-predisposition genes, four of them in DNA repair. REVEL was trained on labelled variants and the AlphaMissense classes were set on ClinVar, so both may overlap our evaluation sets [36, 40]; ESM-2, trained without variant labels, gave similar coverage and validity.

6 Conclusion

CertiVar turns a variant-effect predictor and a few hundred assayed residues of a gene into LoF, functional or no calls with finite-sample guarantees on per-class error rates at a fixed miscoverage level and on call-averaged likelihood ratios with respect to the assay after threshold selection, without an estimate of how

common damaging variants are. Code, data and the supplement (release v1.2.0): `https://github.com/aaryavvatts-lab/certivar`.

References

[1] Melissa J. Landrum, Jennifer M. Lee, Mark Benson, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018.

[2] Vikas Pejaver, Alicia B. Byrne, Bing-Jian Feng, et al. Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *The American Journal of Human Genetics*, 109(12):2163–2177, 2022.

[3] Sue Richards, Nazneen Aziz, Sherri Bale, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*, 17(5):405–424, 2015.

[4] Sean V. Tavtigian, Marc S. Greenblatt, Steven M. Harrison, et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genetics in Medicine*, 20(9):1054–1060, 2018.

[5] Sarah E. Brnich, Ahmad N. Abou Tayoun, Fergus J. Couch, et al. Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework. *Genome Medicine*, 12:3, 2020.

[6] Timothy Bergquist, Sarah L. Stenton, Emily A. W. Nadeau, et al. Calibration of additional computational tools expands ClinGen recommendation options for variant classification with PP3/BP4 criteria. *Genetics in Medicine*, 27(6):101402, 2025.

[7] Mihaly Badonyi and Joseph A. Marsh. acmgscaler: an R package and Colab for standardized gene-level variant effect score calibration within the ACMG/AMP framework. *Bioinformatics*, 41(10):btaf503, 2025.

[8] Shawn Fayer, Carrie Horton, Jennifer N. Dines, et al. Closing the gap: Systematic integration of multiplexed functional data resolves variants of uncertain significance in BRCA1, TP53, and PTEN. *The American Journal of Human Genetics*, 108(12):2248–2258, 2021.

[9] Daniel Zeiberg, Ross Stewart, Shantanu Jain, et al. Gene-based calibration of high-throughput functional assays for clinical variant classification. *bioRxiv*, 2025. doi: 10.1101/2025.04.29.651326.

[10] Malvika Tejura, Shawn Fayer, Abbye E. McEwen, et al. Calibration of variant effect predictors on genome-wide data masks heterogeneous performance across genes. *The American Journal of Human Genetics*, 111(9):2031–2043, 2024.

[11] Yile Chen, Shawn Fayer, Shantanu Jain, et al. Gene- and domain-aware calibration increases the clinical utility of variant effect predictors. *bioRxiv*, 2026. doi: 10.64898/2026.02.17.706269.

[12] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005. doi: 10.1007/b106715.

[13] Mauricio Sadinle, Jing Lei, and Larry Wasserman. Least ambiguous set-valued classifiers with bounded error levels. *Journal of the American Statistical Association*, 114(525):223–234, 2019.

[14] Vladimir Vovk. Conditional validity of inductive conformal predictors. *Machine Learning*, 92(2–3): 349–376, 2013.

[15] Xin Tong, Yang Feng, and Jingyi Jessica Li. Neyman-Pearson classification algorithms and NP receiver operating characteristics. *Science Advances*, 4(2):eaao1659, 2018.

[16] Robin Dunn, Larry Wasserman, and Aaditya Ramdas. Distribution-free prediction sets for two-layer hierarchical models. *Journal of the American Statistical Association*, 118(544):2491–2502, 2023.

[17] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.

[18] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017.

[19] Stephen Bates, Anastasios Angelopoulos, Lihua Lei, et al. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6):43:1–43:34, 2021.

[20] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, et al. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025.

[21] Saad Harrizi, Imane Nait Irahal, Kabine Mostafa, et al. Gene identity, not variant effect, dominates ClinVar benchmarks of missense pathogenicity predictors. *bioRxiv*, 2026. doi: 10.64898/2026.08.21.746180.

[22] D. Katz, J. Baptista, S. P. Azen, et al. Obtaining confidence intervals for the risk ratio in cohort studies. *Biometrics*, 34(3):469–474, 1978.

[23] P. A. R. Koopman. Confidence intervals for the ratio of two binomial proportions. *Biometrics*, 40(2): 513–517, 1984.

[24] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002.

[25] Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3122–3130, 2018.

[26] Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free uncertainty quantification for classification under label shift. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 844–853, 2021.

[27] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, Cambridge, MA, 1999.

[28] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

[29] Gregory M. Findlay, Riza M. Daza, Beth Martin, et al. Accurate classification of BRCA1 variants with saturation genome editing. *Nature*, 562(7726):217–222, 2018.

[30] Huaizhi Huang, Chunling Hu, Jie Na, et al. Functional evaluation and clinical classification of BRCA2 variants. *Nature*, 638(8050):528–537, 2025.

[31] Andrew J. Waters, Timothy Brendler-Spaeth, Danielle Smith, et al. Saturation genome editing of BAP1 functionally classifies somatic and germline variants. *Nature Genetics*, 56(7):1434–1445, 2024.

[32] Rebeca Olvera-León, Fang Zhang, Victoria Offord, et al. High-resolution functional mapping of RAD51C by saturation genome editing. *Cell*, 187(20):5719–5734.e19, 2024.

[33] Xiaoyan Jia, Bala Bharathi Burugula, Victor Chen, et al. Massively parallel functional testing of MSH2 missense variants conferring Lynch syndrome risk. *The American Journal of Human Genetics*, 108(1): 163–175, 2021.

[34] William McLaren, Laurent Gil, Sarah E. Hunt, et al. The Ensembl Variant Effect Predictor. *Genome Biology*, 17(1):122, 2016.

[35] Jun Cheng, Guido Novati, Joshua Pan, et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, 381(6664):eadg7492, 2023.

[36] Nilah M. Ioannidis, Joseph H. Rothstein, Vikas Pejaver, et al. REVEL: an ensemble method for predicting the pathogenicity of rare missense variants. *The American Journal of Human Genetics*, 99 (4):877–885, 2016.

[37] Zeming Lin, Halil Akin, Roshan Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.

[38] Joshua Meier, Roshan Rao, Robert Verkuil, et al. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.

[39] Daniel Esposito, Jochen Weile, Jay Shendure, et al. MaveDB: an open-source platform to distribute and interpret data from multiplexed assays of variant effect. *Genome Biology*, 20(1):223, 2019.

[40] Dominik G. Grimm, Chloé-Agathe Azencott, Fabian Aicheler, et al. The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation*, 36(5): 513–523, 2015.